

AI Assisted Coding

ASSIGNMENT 18.1

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Batch: 31

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

Prompt:

```
# Generate a python code for fetching weather data  
  
# From Open-meteo API and display it in a user-friendly format.  
  
# Use GeoCoding API to get the latitude and longitude of a city entered by the user.  
  
# If API Works Show Weather Data Else Show Error Message Like "Service Temporary Unavailable"  
  
# Handle The Errors Gracefully and Inform the User About It.
```

Code:

```
import requests
```

```
def get_coordinates(city):

    try:

        url = f"https://geocoding-api.open-meteo.com/v1/search?name={city}"

        response = requests.get(url, timeout=5)

        response.raise_for_status()

        data = response.json()

        if data.get('results') and len(data['results']) > 0:

            return data['results'][0]['latitude'], data['results'][0]['longitude']

        else:

            print("City not found.")

            return None

    except requests.exceptions.Timeout:

        print("Request timed out. Please try again.")

        return None

    except requests.exceptions.ConnectionError:

        print("Connection error. Please check your internet connection.")

        return None

    except requests.exceptions.HTTPError:

        print("Service Temporary Unavailable")

        return None

    except ValueError:

        print("Invalid response format.")

        return None

    except Exception as e:

        print(f"An unexpected error occurred: {e}")

        return None


def get_weather(latitude, longitude):
```

```

try:

    url = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&current_weather=true"

    response = requests.get(url, timeout=5)

    response.raise_for_status()

    data = response.json()

    return data['current_weather']

except requests.exceptions.Timeout:

    print("Request timed out. Please try again.")

    return None

except requests.exceptions.ConnectionError:

    print("Connection error. Please check your internet connection.")

    return None

except requests.exceptions.HTTPError:

    print("Service Temporary Unavailable")

    return None

except ValueError:

    print("Invalid response format.")

    return None

except Exception as e:

    print(f"An unexpected error occurred: {e}")

    return None


def main():

    try:

        city = input("Enter the city name: ").strip()

        if not city:

            print("City name cannot be empty.")

```

```

        return

coordinates = get_coordinates(city)

if coordinates:

    weather = get_weather(*coordinates)

    if weather:

        print(f"\nCurrent weather in {city}:")

        print(f"Temperature: {weather['temperature']}°C")

        print(f"Wind Speed: {weather['windspeed']} km/h")

        print(f"Weather Code: {weather['weathercode']}")

    else:

        print("Failed to retrieve weather data.")

except KeyboardInterrupt:

    print("\nProgram interrupted by user.")

except Exception as e:

    print(f"An unexpected error occurred: {e}")


if __name__ == "__main__":

    main()

```

Output:

Enter the city name: Warangal

Current weather in Warangal:

Temperature: 29.2°C

Wind Speed: 3.3 km/h

Weather Code: 3

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

Prompt:

```
# Generate a python code for converting INR to other currencies
# Use a free currency exchange API
# Take amount as input from user
# Convert INR to USD, EUR, and GBP
# Display output in a clear table format
# If API fails show message "Service Temporary Unavailable"
# Handle invalid input and errors properly
```

Code:

```
import requests

def get_exchange_rates():

    try:

        url = "https://api.exchangerate-api.com/v4/latest/INR"

        response = requests.get(url, timeout=5)

        response.raise_for_status()

        data = response.json()
```

```

        return data.get('rates', {})

except requests.exceptions.Timeout:

    print("Request timed out. Please try again.")

    return None

except requests.exceptions.ConnectionError:

    print("Connection error. Please check your internet connection.")

    return None

except requests.exceptions.HTTPError:

    print("Service Temporary Unavailable")

    return None

except ValueError:

    print("Invalid response format.")

    return None

except Exception as e:

    print(f"An unexpected error occurred: {e}")

    return None

def convert_inr(amount, rates):

    try:

        usd = amount * rates.get('USD', 0)

        eur = amount * rates.get('EUR', 0)

        gbp = amount * rates.get('GBP', 0)

        return usd, eur, gbp

    except Exception as e:

        print(f"An error occurred during conversion: {e}")

        return None

def main():

    try:

        amount_str = input("Enter the amount in INR: ").strip()

```

```

if not amount_str.replace('.', '', 1).isdigit():

    print("Invalid input. Please enter a numeric value.")

    return

amount = float(amount_str)

rates = get_exchange_rates()

if rates:

    usd, eur, gbp = convert_inr(amount, rates)

    print(f"{'Currency':<10} {'Amount':>10}")

    print(f"{'USD':<10} {usd:>10.2f}")

    print(f"{'EUR':<10} {eur:>10.2f}")

    print(f"{'GBP':<10} {gbp:>10.2f}")

except Exception as e:

    print(f"An unexpected error occurred: {e}")

if __name__ == "__main__":

    main()

```

Output:

```

Enter the amount in INR: 1000
Currency      Amount
USD           10.60
EUR           9.19
GBP           7.95

```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

Prompt:

```
# Generate a python code for fetching repository details
# Use a free API to get repository information
# Take repository name as input (owner/repo)
# Display name, description, stars, forks, and open issues
# If repository not found show proper message
# Handle API errors and invalid input
```

Code:

```
import requests

def get_repository_info(repo_name):

    try:

        url = f"https://api.github.com/repos/{repo_name}"

        response = requests.get(url, timeout=5)
```

```

if response.status_code == 404:

    print("Repository not found.")

    return None

response.raise_for_status()

data = response.json()

name = data.get('name', 'N/A')

description = data.get('description', 'N/A')

stars = data.get('stargazers_count', 'N/A')

forks = data.get('forks_count', 'N/A')

open_issues = data.get('open_issues_count', 'N/A')

return name, description, stars, forks, open_issues

except requests.exceptions.Timeout:

    print("Request timed out. Please try again.")

    return None

except requests.exceptions.ConnectionError:

    print("Connection error. Please check your internet connection.")

    return None

except requests.exceptions.HTTPError as e:

    status_code = e.response.status_code if e.response is not None else 'unknown'

    print(f"HTTP error while contacting GitHub API: {status_code}")

    return None

except ValueError:

    print("Invalid response format.")

    return None

except Exception as e:

    print(f"An unexpected error occurred: {e}")

    return None

def main():

```

```
try:

    repo_name = input("Enter the repository name (owner/repo): ").strip()

    if '/' not in repo_name:

        print("Invalid input. Please enter in the format 'owner/repo'.")

        return

    info = get_repository_info(repo_name)

    if info:

        name, description, stars, forks, open_issues = info

        print(f"Name: {name}")

        print(f"Description: {description}")

        print(f"Stars: {stars}")

        print(f"Forks: {forks}")

        print(f"Open Issues: {open_issues}")

    except Exception as e:

        print(f"An unexpected error occurred: {e}")

if __name__ == "__main__":

    main()
```

Output:

```
Enter the repository name (owner/repo): FreeCodeCamp/freecodecamp
Name: freeCodeCamp
Description: freeCodeCamp.org's open-source codebase and curriculum. Learn math, programming, and computer science for free.
Stars: 438816
Forks: 43768
Open Issues: 225
```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).
- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling

Prompt:

```
# Generate a python code for fetching news headlines
# Use https://newsapi.org/ API
# Take category as input (sports, technology, health)
# Display top 5 headlines in numbered format
# If API fails show message "Service Temporary Unavailable"
# Handle invalid category and errors properly
# Retry once if request fails
```

Code:

```
import requests

def get_exchange_rates():

    try:

        url = "https://api.exchangerate-api.com/v4/latest/INR"

        response = requests.get(url, timeout=5)

        response.raise_for_status()
```

```

    data = response.json()

    return data.get('rates', {})

except requests.exceptions.Timeout:

    print("Request timed out. Please try again.")

    return None

except requests.exceptions.ConnectionError:

    print("Connection error. Please check your internet connection.")

    return None

except requests.exceptions.HTTPError:

    print("Service Temporary Unavailable")

    return None

except ValueError:

    print("Invalid response format.")

    return None

except Exception as e:

    print(f"An unexpected error occurred: {e}")

    return None

def convert_inr(amount, rates):

    try:

        usd = amount * rates.get('USD', 0)

        eur = amount * rates.get('EUR', 0)

        gbp = amount * rates.get('GBP', 0)

        return usd, eur, gbp

    except Exception as e:

        print(f"An error occurred during conversion: {e}")

        return None

def main():

    try:

```

```

amount_str = input("Enter the amount in INR: ").strip()

if not amount_str.replace('.', '', 1).isdigit():

    print("Invalid input. Please enter a numeric value.")

    return

amount = float(amount_str)

rates = get_exchange_rates()

if rates:

    usd, eur, gbp = convert_inr(amount, rates)

    print(f"{'Currency':<10} {'Amount':>10}")

    print(f"{'USD':<10} {usd:>10.2f}")

    print(f"{'EUR':<10} {eur:>10.2f}")

    print(f"{'GBP':<10} {gbp:>10.2f}")

except Exception as e:

    print(f"An unexpected error occurred: {e}")

if __name__ == "__main__":

    main()

```

Output:

```

Enter news category (sports, technology, health): health
1. The surprising link between back pain and a sensitivity to loud noises - The Washington Post (Source: The Washington Post, Published At: 2026-03-24T14:22:24Z)
2. Exercise can lower Alzheimer's risk. Scientists may have discovered why. - The Washington Post (Source: The Washington Post, Published At: 2026-03-24T10:32:01Z)
3. Woman, 41, Was Tired of 'Hiding' Excess Skin After losing 180 lbs. on a GLP-1. Then She Got an 'Awake' Arm Lift (Exclusive) - Yahoo Life UK (Source: PEOPLE, Published At: 2026-03-24T08:30:00Z)
4. 2 Utah companies are quietly changing how AI actually works for real businesses - KSL.com (Source: KSL.com, Published At: 2026-03-24T03:02:25Z)
5. Psychology says people who describe themselves as joyful after 50 didn't suddenly become optimistic - they stopped treating happiness like a reward for perfect behavior and started treating it li
ke a practice - Vegout (Source: Vegoutmag.com, Published At: 2026-03-24T01:45:21Z)

```